TABLE 1
BILL OF MATERIAL FOR TRANSMITTER

Quantity	Item/Model	Description
1	TP4056 (MOD1)	TP4056 on a breakout board
1	Battery cell	18650 3.7V lithium-ion rechargeable battery
1	SW_SPST (SW1)	Switch for input power
1	ESP32 DEVKITC-32D (MOD2)	ESP32 development board
1	MPU6050 (MOD3)	MPU6050 3-axis gyroscope and accelerometer

TABLE 2
BILL OF MATERIAL FOR RECEIVER

Quantity	Item	Description
2	Screw_terminal_01x02	Screw terminal block for input and output
1	AC to DC 5V power	5V AC to DC power
2	1N4007 (D1)	General-purpose diode
1	1k (R1)	¼W through-hole resistor
1	BC547 (T1)	NPN transistor
1	ESP32 DEVKITC-32D (MOD1)	ESP32 development board
1	RAYEX-L90 (RL1)	30A T-type relay

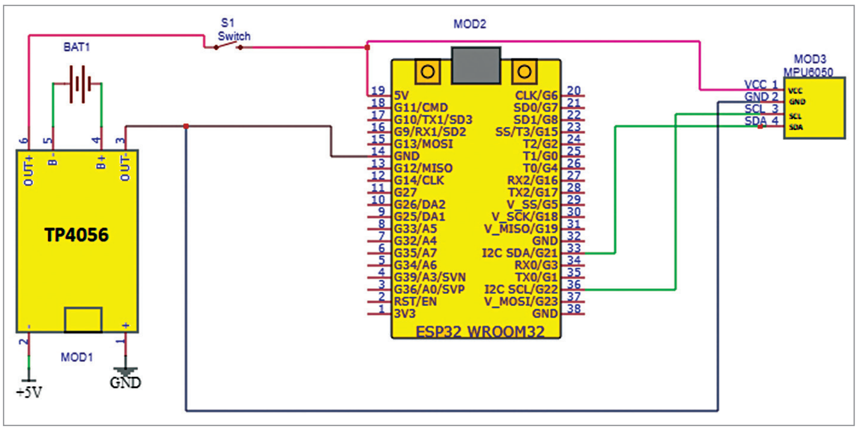

Fig. 4: Circuit diagram of the transmitter

processes the data and sends it to the receiver using ESP-NOW protocol. The transmitter uses 3.7V lithium-ion rechargeable battery, which is charged through TP4056 charging module that is connected to a 5V USB charger.

The receiver receives the command, if the command is valid (either 0 or 1), and triggers the relay. The ESP32 microcontroller transceiver is used for getting valid commands and to trigger the relay. The receiver uses 230V AC to 5V DC converter and a 5V relay module to switch the iron on or off. Whenever a valid command is received, ESP32 triggers a 'high' or 'low' signal to the GPIO pins connected to the relay, which switches the iron on/off.

The circuit diagram of sensing part (transmitter) is shown in Fig. 4. The transmitter is built around TP4056 (MOD1), ESP32 (MOD2), and MPU6050 (MOD3).

The circuit diagram of triggering part (receiver) is shown in Fig. 5. The receiver is built around ESP32 DEVKITC-32D (MOD1), AC to DC 5V power source (PS1), and 5V single-changeover relay.

As the project has two parts, we need a protocol for communication between them. The ESP-NOW wireless communication protocol used for the purpose enables direct,

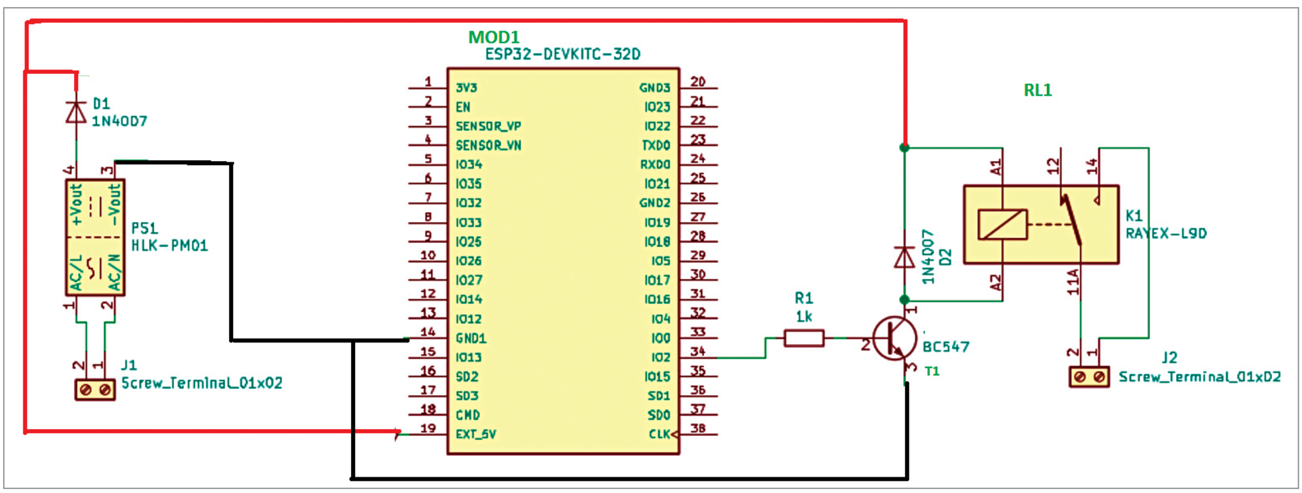

Fig. 5: Circuit diagram of the receiver